

Classical Statistical Mechanics

Half-baked notes from PHYS 35200

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Teaching thermal physics is as easy as a song:
you think you make it simpler when you make it slightly wrong.

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This is a *heavily* processed version of notes that I took during UChicago’s graduate-level statistical mechanics course in Spring 2025 with Professor Dam Son, covering classical and quantum statistical mechanics. This began to be crafted in order to be used for the open-note final, but this ended up not being necessary as the final was changed to being take-home. Therefore, it does not include all of the content in the class.

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1 Principles of Statistical Mechanics

1.1 The distribution function

In every class before statistical mechanics, the methods for solving physics problems were basically the same: write down the equations of motion, plug in known values for the particles in your system and solve. This works fine for one or two particles. However, if you want to describe a typical thermodynamic system with $\sim 10^{23}$ particles, it quickly becomes impractical to write down the equations of motion for each particle, let alone solve. We need the tools of *statistical mechanics* to solve these problems.

Instead of focusing on one particular point in phase space and its trajectory, imagine we have some *ensemble* of trajectories. As we increase the number of points, they effectively form a continuous distribution known as the *distribution function*.

Definition 1.1. The *distribution function* is defined as

$$\rho(q_1, \dots, q_n, p_1, \dots, p_n) = \lim_{\Delta V \rightarrow 0} \frac{\{\# \text{ of points in volume}\}}{\{\text{total } \# \text{ of points}\} \Delta V}$$

Since ρ is basically a probability distribution, we require the normalization condition

$$\int d^N q d^N p \rho(q, p) = 1$$

1.2 Ergodicity

fill in after midterm

1.3 Liouville's theorem

Once we have our distribution function, we can do some cool things with it.

Theorem 1.1. *Liouville's theorem:*

$$\frac{d\rho(q, p, t)}{dt} = \frac{\partial \rho}{\partial t} - \{H, \rho\} = 0$$

In other words, the density of the ensemble along a trajectory does not change in time.

For those very familiar with classical mechanics, this follows basically immediately from time evolution obeying the symplectic condition. Here is an alternative proof.

Proof. In the case of one coordinate, consider a box of volume $\Delta q_i \times \Delta p_i$ in phase space. For a sufficiently small time interval Δt , particles can only enter or leave the box if they are sufficiently close to one of the sides. Observe that in the q direction we have a net flux of approximately

$$\rho\left(p + \frac{\Delta p}{2}, q\right) \dot{q} \Delta t \Delta p - \rho\left(p + \frac{\Delta p}{2}, q + \Delta q\right) \dot{q} \Delta t \Delta p$$

and in the p direction we have a net flux of

$$\rho\left(p, q + \frac{\Delta q}{2}\right) \dot{p} \Delta t \Delta q - \rho\left(p + \Delta p, q + \frac{\Delta q}{2}\right) \dot{p} \Delta t \Delta q$$

hence

$$\begin{aligned} \frac{\Phi}{\Delta t \Delta p \Delta q} \approx & \frac{\rho\left(p + \frac{\Delta p}{2}, q\right) - \rho\left(p + \frac{\Delta p}{2}, q + \Delta q\right)}{\Delta q} \dot{q} \\ & + \frac{\rho\left(p, q + \frac{\Delta q}{2}\right) - \rho\left(p + \Delta p, q + \frac{\Delta q}{2}\right)}{\Delta p} \dot{p} \end{aligned}$$

Taking the limit as $\Delta t, \Delta p, \Delta q$ go to 0 and recalling that

$$\dot{p} = \frac{\partial H}{\partial q} \quad \dot{q} = -\frac{\partial H}{\partial p}$$

we see that

$$\frac{\partial \rho}{\partial t} = -\frac{\partial \rho}{\partial q} \frac{\partial H}{\partial p} + \frac{\partial \rho}{\partial p} \frac{\partial H}{\partial q} = \{H, \rho\}$$

and this generalizes to the case of multiple coordinates simply by adding more terms to Φ . $\square$

1.4 Density matrices

We can describe ensembles of classical particles with a probability distribution over phase space. How do we describe ensembles of *quantum* particles? We do so with *density matrices* (or *operators*). [fill in explanation of density matrices after midterm].

The time evolution of a density matrix turns out to be exactly the result we'd expect from "naive" canonical quantization of Liouville's theorem.

$$\frac{\partial \hat{\rho}}{\partial t} = -\frac{i}{\hbar} [H, \hat{\rho}]$$

2 The Microcanonical Ensemble

2.1 “Easy to explain, impossible to prove”

A lot of things in the world seem to be in *thermal equilibrium*; that is, their thermodynamic properties don’t change much over time. Why?

This turns out to be a very tricky question to answer mathematically in the general case. However, physicists can do better, since while mathematics has theorems, physics has assumptions.¹ Therefore, let us declare the following to be true:

- There exists a thermal equilibrium; that is, $\exists \rho$ such that $\frac{\partial \rho}{\partial t} = 0$
- Moreover, ρ is described by a *microcanonical ensemble*: for some specified E ,

$$\rho(q, p) = \begin{cases} \text{const.} & E < H(q, p) < E + \Delta E \\ 0 & \text{otherwise} \end{cases}$$

Observe that any ρ that is solely a function of H satisfies

$$\frac{\partial \rho}{\partial t} = \{H, \rho\} = \sum_i \left(\frac{\partial H}{\partial q_i} \left(\frac{\partial \rho}{\partial H} \frac{\partial H}{\partial p_i} \right) - \frac{\partial H}{\partial p_i} \left(\frac{\partial \rho}{\partial H} \frac{\partial H}{\partial q_i} \right) \right) = 0$$

Therefore, once we’ve specified the energy of the system E , ρ is a state of thermal equilibrium at energy E . We can also consider the quantum version of the microcanonical ensemble, where we define our equilibrium density matrix as

$$\hat{\rho} = \sum_i p_i |E_i\rangle \langle E_i|$$

where

$$p_i = \begin{cases} \text{const.} & E_i \in [E, E + \Delta E] \\ 0 & \text{otherwise} \end{cases}$$

We can work out this constant by defining

Definition 2.1. The *number of microstates* Ω is the number of energy eigenstates with energy between E and $E + \Delta E$.

¹Spherical cow in a vacuum...

At high energies, we worked out a general relationship in homework between Ω for systems where the WKB approximation is valid and the volume of valid energies in the corresponding classical phase space.

$$\Omega = \frac{1}{(2\pi\hbar)^N} V_{\text{classical}}$$

This lets us define Ω for classical systems as well.

2.2 Entropy and the 2nd Law of Thermodynamics

One interesting quantity is the *entropy*.

Definition 2.2. The *entropy* S of a system is equal to $k_B \ln(\Omega)$.

Here, k_B is Boltzmann's constant. Professor Son uses the convention $k_B = 1$, so it will not appear in the rest of these notes.

In principle, S depends both on E and ΔE . However, assuming a continuous enough distribution of energy eigenstates, we have that a rescaling $\Delta E \mapsto 2\Delta E$ maps $\Omega \mapsto 2\Omega$ and $S \mapsto S + \ln(2)$, and the constant factor ends up not really mattering.

One of the reasons we care about entropy is that it is *extensive*. In many systems, $\Omega \sim e^N$, so $S \sim N$ scales nicely with system size. Additionally, for two isolated systems,

$$\Omega(E_1, E_2) = \Omega_1(E_1)\Omega_2(E_2) \implies S(E_1, E_2) = S_1(E_1) + S_2(E_2)$$

so the total entropy is *additive*.

The far more important reason to care about entropy is the *second law of thermodynamics*: in a closed system, S is always increasing. The intuition for this is that S measures the “likelihood” of a system being in a certain state, and systems will never move from a more likely state to a less likely state.

2.3 Temperature

Definition 2.3. The *temperature* T of a thermodynamic system is defined as

$$T = \frac{1}{\frac{\partial S}{\partial E}}$$

It's not apparent how this definition connects to the “average kinetic energy” explanation we got in elementary school. In general, it doesn't! See (6.4) for the connection.

One thing that *is* apparent is that this definition fixes the flow of energy. Assuming a change in energy between two systems

$$E_1 \mapsto E_1 + \delta E \quad E_2 \mapsto E_2 - \delta E$$

the Second Law of Thermodynamics demands

$$\delta E \left(\frac{1}{T_1} - \frac{1}{T_2} \right) > 0$$

Therefore, energy only spontaneously flows from hot to cold things and not the other way around.

2.4 Examples

2.4.1 Two-level ensemble

Consider a system of N two-level systems with energy gap ϵ . The energy is given by

$$E = \epsilon N_E$$

where N_E is the number of systems in the excited state. A simple combinatorics argument gives

$$\Omega = \binom{N}{N_E} = \frac{N!}{N_E!(N - N_E)!}$$

$$S = \ln(N!) - \ln(N_E!) - \ln((N - N_E)!)$$

Using *Stirling's approximation* $\ln(N!) \approx N \ln(N) - N$ we get that

$$\begin{aligned} S &\approx N \ln(N) - N - N_E \ln(N_E) + N_E - (N - N_E) \ln(N - N_E) + (N - N_E) \\ &= N \ln(N) - N_E \ln(N_E) - (N - N_E) \ln(N - N_E) \end{aligned}$$

Defining $x = \frac{N_E}{N}$, we see that

$$\begin{aligned} S &= N(\ln(N) - x \ln(Nx) - (1 - x) \ln(N(1 - x))) \\ &= N(-x \ln(x) - (1 - x) \ln(1 - x)) \end{aligned}$$

Plotting S against x (which is proportional to the energy), we observe something striking: the model has infinite temperature for $x = \frac{1}{2}$ and negative temperature for $x > \frac{1}{2}$. This is a real effect that has been observed in certain systems e.g. lasers. However, this model isn't too realistic for most situations we encounter, since at high energies we are frequently able to excite systems beyond their first excited state.

2.4.2 Harmonic oscillator ensemble

Consider a system of N quantum harmonic oscillators. The energy is given by

$$E = \hbar\omega \sum_i n_i$$

where n_i is the energy level of the i th QHO (and we ignore the zero-point energy). Using the classic “stars and bars” technique from competition math,² we can compute

$$\Omega = \frac{(N + N_E - 1)!}{(N - 1)!N_E!}$$

where $N_E = \frac{E}{\hbar\omega}$. We can therefore compute

$$S \approx N[(x + 1) \ln(x + 1) - x \ln(x)]$$

where $x = \frac{N_E}{N}$ as above. Note that for this system

$$T = \frac{1}{\frac{\partial S}{\partial E}} \sim \frac{1}{\frac{1}{x}} \sim E$$

which is more intuitive than the previous example.

3 Thermodynamics

3.1 The 1st Law of Thermodynamics

Consider a Hamiltonian that is time-dependent through some parameter λ , i.e. $H = H(q, p, \lambda(t))$. Then

$$\frac{d}{dt} \langle E \rangle = \left\langle \frac{\partial H}{\partial \lambda} \right\rangle \dot{\lambda}$$

Now let's say we have a Hamiltonian of the form

$$H = H_0(x, p) + \sum V(z_i - z_0)$$

²Here I feel it is relevant to point out that Professor Son got a perfect score on the IMO in 1984.

where we start with a generic Hamiltonian in 3D space and impose a (smooth) wall at $z = z_0$. We can write

$$\frac{\partial H}{\partial z_0} = \sum_i \frac{\partial}{\partial z_0} V(z_i - z_0) = - \sum_i \frac{\partial}{\partial z_i} V(z_i - z_0) = F$$

where we identify the last term as the force on the ensemble by the wall. Let us write

$$\langle F \rangle = -P \cdot L_x \cdot L_y$$

using the *pressure* P : the average force on the wall by the ensemble per unit area.

Now, consider a time-dependent $z_0 = \lambda(t)$. Since $\frac{dS}{dt}$ is a function of $\dot{\lambda}$, we can expand

$$\frac{dS}{dt}(\dot{\lambda}) \approx 0 + c_1 \dot{\lambda} + c_2 \dot{\lambda}^2 + \dots$$

The 2nd Law of Thermodynamics demands $c_1 = 0$. Therefore, at best, $\frac{dS}{dt}(\dot{\lambda}) \sim \dot{\lambda}^2$. Realizing that

$$\Delta S = \int_0^T \frac{dS}{dt} dt \sim \tau \cdot \frac{1}{\tau^2} \rightarrow 0 \text{ as } \tau \rightarrow \infty$$

Therefore, if we move the wall slow enough, we get no change in entropy. This is known as an *adiabatic process*. We can then say that

$$\left(\frac{dE}{d\lambda} \right)_S = F$$

Combined with

$$\left(\frac{dE}{dS} \right)_\lambda = T$$

the magic of 1-forms lets us stitch them together into $dE = TdS + Fd\lambda$. Noting that $L_x \cdot L_y \cdot d\lambda = dV$ is the change in *volume* of the system, we obtain the *1st law of thermodynamics*.

$$dE = TdS - PdV$$

3.2 Ideal gas

We are finally ready to study our first real thermodynamic system:³ the *ideal gas*, consisting of non-interacting particles in a box. The Hamiltonian is

$$H = \sum \frac{p_i^2}{2m}$$

As written, this will not reach thermal equilibrium, but this can be explained away by saying there's a negligible but nonzero interaction force. Our procedure is to first find $\Omega(E)$, then S , then solve for E in terms of S and V , and finally solve for thermodynamic variables of interest.

3.2.1 Deriving S

We first need to count the number of microstates. We are going to impose the slightly unrealistic assumption of periodic boundary conditions. This ends up not mattering that much, since it's the behavior of the bulk that matters more. (In homework, we derived the case where the box is in an infinite well, so the wave function is 0 at the boundary). However, it lets us cleanly count the microstates of a single particle: each energy eigenstate for a single particle looks like

$$e^{\frac{i}{\hbar} \vec{p} \cdot \vec{x}}$$

with

$$p_x = \frac{2\pi\hbar}{L_x} n_x \quad p_y = \frac{2\pi\hbar}{L_y} n_y \quad p_z = \frac{2\pi\hbar}{L_z} n_z$$

Let's assume $L_x = L_y = L_z = L$. The energy of the ensemble is then given by

$$E = \sum_{i=1}^{3N} \frac{1}{2m} \left(\frac{2\pi\hbar}{L} \right)^2 n_i^2$$

The number of tuples with energy less than E be approximated by considering the volume of a $3N$ -dimensional sphere of radius $\frac{L}{2\pi\hbar} \sqrt{2mE}$. Therefore, the number of tuples in the microcanonical ensemble is approximately

$$\left(\frac{L}{2\pi\hbar} \right)^{3N} (V_{3N\text{-dimensional sphere}}(\sqrt{2m(E + \Delta E)}) - V_{3N\text{-dimensional sphere}}(\sqrt{2mE}))$$

³As the ancient texts say, "Perhaps it will be wise to approach the subject cautiously."

Is this the number of microstates? No! Remember that in quantum mechanics particles are *indistinguishable*, which means we're overcounting most states by a factor of $N!$. Therefore, our real Ω is

$$\frac{1}{N!} \left(\frac{L}{2\pi\hbar} \right)^{3N} (V_{3N}(\sqrt{2m(E + \Delta E)}) - V_{3N}(\sqrt{2mE}))$$

At sufficiently high energies, the states with duplicate momenta become negligible, so we don't have to worry about casework. The volume term is approximated by

$$(\sqrt{2mE})^{3N-1} S_{3N-1}(1) \cdot \Delta\sqrt{2mE}$$

where $S_{3N-1}(1)$ is the surface area of the $3N$ -dimensional unit ball. There's actually a neat trick for computing this. Using generalized spherical coordinates, observe that

$$\pi^{d/2} = \int_{\mathbb{R}^d} e^{-x_1^2 - \dots - x_d^2} d^d \vec{x} = \int_0^\infty r^{d-1} e^{-r^2} S_{d-1}(1) dr = \frac{1}{2} \Gamma\left(\frac{d}{2}\right) S_{d-1}(1)$$

so

$$S_{d-1}(1) = \frac{2\pi^{d/2}}{\Gamma\left(\frac{d}{2}\right)}$$

Putting this all together, we get that

$$\Omega = \frac{1}{N!} \frac{V^N}{(2\pi\hbar)^{3N}} (\sqrt{2mE})^{3N-1} \frac{2\pi^{3N/2}}{\Gamma\left(\frac{3N}{2}\right)} \frac{\Delta E \sqrt{m}}{\sqrt{2E}}$$

and

$$S = N \ln \left(\frac{1}{(2\pi\hbar)^3} \frac{V}{N} \left(\frac{4\pi m E}{3N} \right)^{3/2} \right) + \frac{5}{2}N + \text{junk}$$

This is the *Sackur–Tetrode equation*. Observe that

$$S(\alpha E, \alpha V, \alpha N) = \alpha S(E, V, N)$$

so S is extensive. This would not have been the case without the indistinguishability of quantum particles!

3.2.2 The Ideal Gas Law

We can solve for E in terms of S and V

$$E = \frac{3N}{4\pi n} (2\pi\hbar)^2 \left(\frac{N}{V}\right)^{2/3} e^{\frac{2S}{3N} - \frac{5}{3}}$$

We can then compute

$$T = \frac{\partial E}{\partial S} = \frac{2}{3N} E \implies E = \frac{3}{2} NT$$

which recovers the elementary school formula. (Remember $k_B = 1$.) We can also compute

$$P = -\frac{\partial E}{\partial V} = \frac{2}{3} \frac{E}{V} = \frac{NT}{V}$$

which recovers the *ideal gas law*

$$PV = NT$$

3.3 Gibbs mixing entropy

Classical thermodynamics had an issue with the following problem: suppose we have two containers of the same gas with a divider between them. We can see from the Sackur-Tetrode equation (or from the Maxwell relations in 4.2) that

$$\left(\frac{\partial S}{\partial V}\right)_T = \frac{N}{V}$$

If we open the divider, the theory would predict a change in entropy

$$\Delta S = N_1 \ln \left(\frac{V_1 + V_2}{V_1}\right) + N_2 \ln \left(\frac{V_1 + V_2}{V_2}\right)$$

However, if you then close the divider, the system returns to its original state, violating the 2nd law of thermodynamics. What gives?

If the gases were distinguishable, then there is a real change in entropy, but closing the divider would not return the system to its original state. What we are forgetting is the indistinguishability of quantum particles. The classical formula doesn't take into account that when the gases are indistinguishable, N changes, so the number of duplicate microstates goes from $N_1!N_2!$ to $(N_1 + N_2)!$. If we properly use an extensive formula for entropy like the Sackur-Tetrode equation, then $\Delta S = 0$ and there is no paradox.

4 Thermodynamic potentials

4.1 E , H , F and G

Starting from E , we can derive some more interesting quantities using Legendre transforms.

Name	Value	Differential
Enthalpy	$H = E + PV$	$dH = TdS + VdP$
Helmholtz free energy	$F = E - TS$	$dF = -SdT - PdV$
Gibbs free energy	$G = E - TS + PV$	$dG = -SdT + VdP$

4.2 Maxwell relations

With these potentials, we can derive the *Maxwell relations*.

Maxwell's relations(common)

$$\begin{aligned}
 +\left(\frac{\partial T}{\partial V}\right)_S &= -\left(\frac{\partial P}{\partial S}\right)_V = \frac{\partial^2 U}{\partial S \partial V} \\
 +\left(\frac{\partial T}{\partial P}\right)_S &= +\left(\frac{\partial V}{\partial S}\right)_P = \frac{\partial^2 H}{\partial S \partial P} \\
 +\left(\frac{\partial S}{\partial V}\right)_T &= +\left(\frac{\partial P}{\partial T}\right)_V = -\frac{\partial^2 F}{\partial T \partial V} \\
 -\left(\frac{\partial S}{\partial P}\right)_T &= +\left(\frac{\partial V}{\partial T}\right)_P = \frac{\partial^2 G}{\partial T \partial P}
 \end{aligned}$$

Figure 1: Note: $U = E$

These are sometimes lumped with three purely mathematical facts

$$\begin{aligned}
 \left(\frac{\partial y}{\partial x}\right)_z &= \left(\left(\frac{\partial x}{\partial y}\right)_z\right)^{-1} \\
 \left(\frac{\partial w}{\partial y}\right)_z &= \left(\frac{\partial w}{\partial x}\right)_z \left(\frac{\partial x}{\partial y}\right)_z \\
 -1 &= \left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y
 \end{aligned}$$

Note the minus sign in the last one!

4.3 Spontaneous processes

Here we explore the utility of the other thermodynamic potentials. Suppose we have a process involving a change in both entropy ΔS and energy ΔE . The 2nd Law says that $\Delta S \geq 0$ in an isolated system. Suppose our system is not isolated, but attached to a thermal bath at positive temperature T . Then

$$\Delta S_{\text{total}} = \Delta S - \frac{1}{T}\Delta E = -\frac{1}{T}(\Delta E - T\Delta S) = -\frac{\Delta F}{T}$$

Hence, the process is spontaneous if and only if $\Delta F \leq 0$. Now suppose there is a process involving a change of entropy ΔS , energy ΔE , and volume ΔV , at constant T and P . (This is often the case in chemistry.) Then

$$\Delta S_{\text{total}} = \Delta S - \frac{1}{T}(\Delta E + P\Delta V) = -\frac{\Delta G}{T}$$

Therefore, the process is spontaneous if and only if $\Delta G \leq 0$.

4.4 Applications

4.4.1 Unmixing air

Suppose we have a $5L$ box containing $4L$ of N_2 and $1L$ of O_2 . How much work will it take to separate the gases? For the process to occur, we need $\Delta F = \Delta E - T\Delta S \leq 0$. We know that to reverse the Gibbs mixing entropy we have

$$\Delta S = -N_1 \ln \left(\frac{V_1 + V_2}{V_1} \right) - N_2 \ln \left(\frac{V_1 + V_2}{V_1} \right)$$

Therefore, the amount of work that needs to be done is

$$-\Delta E \geq TN_1 \ln \left(\frac{V_1 + V_2}{V_1} \right) + TN_2 \ln \left(\frac{V_1 + V_2}{V_1} \right)$$

Observe also for ideal gases $PV = TN$, making the work more apparent.

4.4.2 Defining thermodynamic quantities

We can define

$$C_v = \left(\frac{\partial E}{\partial T} \right)_V = T \left(\frac{\partial S}{\partial T} \right)_V \quad C_p = T \left(\frac{\partial S}{\partial T} \right)_P = \left(\frac{\partial H}{\partial T} \right)_P$$

the *heat capacity at constant volume* and *constant pressure*, respectively. Additionally, there's the thermal expansion coefficient

$$\alpha = \frac{1}{V} \left(\frac{\partial V}{\partial T} \right)_P$$

and *isothermal compressibility*

$$k = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_T$$

These are not interesting theoretically but are defined since they are useful in engineering.

4.4.3 Heating up a box

Suppose we are heating up a box and measuring its pressure. We can compute

$$\left(\frac{\partial P}{\partial T} \right)_V = - \left(\frac{\partial P}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P = - \frac{\left(\frac{\partial V}{\partial T} \right)_P}{\left(\frac{\partial V}{\partial P} \right)_T} = \frac{\alpha V}{k V} = \frac{\alpha}{k}$$

4.4.4 Joule free expansion

Suppose we allow a gas to expand suddenly $V \rightarrow \Delta V$ with no change in E . Then

$$\begin{aligned} \left(\frac{\partial T}{\partial V} \right)_E &= - \frac{\left(\frac{\partial E}{\partial V} \right)_T}{\left(\frac{\partial E}{\partial T} \right)_V} = - \frac{1}{C_v} \left(\frac{\partial E}{\partial V} \right)_T \\ &= - \frac{1}{C_v} \left(T \left(\frac{\partial S}{\partial V} \right)_T - P \right) = - \frac{1}{C_v} \left(T \left(\frac{\partial P}{\partial T} \right)_V - P \right) \end{aligned}$$

Observe that for an ideal gas this quantity equals 0.

5 The Variable N

5.1 Chemical potential

What happens when we let the number of particles in our system vary? Writing our energy $E = E(S, V, N)$, our 1st law becomes

$$dE = TdS - PdV + \mu dN$$

where $\mu := \left(\frac{\partial E}{\partial N} \right)_{S,V}$ is the *chemical potential*.

Something to note is that, assuming our system is nice enough, the terms in the 1st law always pair an *extensive* quantity (something that scales linearly with system size, like N, S, V) with an *intensive quantity* (something that doesn't, like T, P, μ) to form an extensive quantity (our thermodynamic potentials E, H, F, G). In the case of μ , “nice enough” means no significant long-range interactions.

If we're clever enough, we can notice that since the Gibbs free energy $G(T, P, N)$ is only a function of one extensive variable, we have that

$$G(T, P, \lambda N) = \lambda G(T, P, N) \implies G = NG(T, P, 1)$$

Since $\mu = \left(\frac{\partial G}{\partial N}\right)_{T,P}$, we get immediately that

$$\mu = \frac{G}{N}$$

That is, the chemical potential is exactly the Gibbs free energy per particle.

5.2 Grand potential

We can define a new thermodynamic potential, the *grand potential*⁴

$$\Omega = E - TS - \mu N \quad d\Omega = -SdT - PdV - Nd\mu$$

Using the same trick as above, we get that

$$\omega(T, \mu) := \frac{\Omega}{V} = -P$$

and the corresponding differential form

$$d\Omega = V \left(\frac{\partial \omega}{\partial T} \right)_{\mu} dT + \omega dV + V \left(\frac{\partial \omega}{\partial \mu} \right)_{\mu} d\mu$$

gives us the new relations

$$V \left(\frac{\partial P}{\partial T} \right)_{\mu} = S \quad V \left(\frac{\partial P}{\partial \mu} \right)_{\mu} = N$$

⁴Honestly, I think it's just the “pretty good” potential

5.3 External fields

Suppose we have system in an external field such that its Hamiltonian looks like

$$H = \sum_{i=1}^N \frac{p_i^2}{2m} + \sum_{\langle ij \rangle} V(\vec{x}_i - \vec{x}_j) + \sum_{i=1}^N U(\vec{x}_i)$$

Lets imagine discretizing the potential into a bunch of small bins and assuming the potential is an approximately constant U_i in each bin. The equilibrium configuration will be the one that maximizes the total entropy

$$S = \sum S_i(E_i, N_i)$$

subject to the constraints

$$N = \sum_i N_i \quad E = \sum_i (E_i + U_i N_i)$$

where E_i is the energy without the field. we can solve this with our favorite constrained optimization technique: Lagrange multipliers! Writing

$$\tilde{S}(E_i, N_i, \lambda_1, \lambda_2) = \sum_i S_i(E_i, N_i) - \lambda_1 \left(\sum_i (E_i + U_i N_i) - E \right) - \lambda_2 \left(\sum_i N_i - N \right)$$

we observe that for the equilibrium configuration

$$\frac{\partial \tilde{S}}{\partial E_i} = 0 \implies \frac{1}{T_i} = \lambda_1$$

so all of the bins are at the same temperature, as expected. Likewise,

$$\frac{\partial \tilde{S}}{\partial N_i} = 0 \implies \frac{\partial S_i}{\partial N_i} - \lambda_1 U_i - \lambda_2 = 0$$

Using the relation

$$\left(\frac{\partial S_i}{\partial N_i} \right)_{E_i} = -1 \cdot \left(\frac{\partial S_i}{\partial E_i} \right)_{N_i} \left(\frac{\partial E_i}{\partial N_i} \right)_{S_i}$$

we get that

$$-\frac{\mu_i}{T} - \frac{U_i}{T} - \lambda_2 = 0 \implies \mu(x) + U(x) \text{ is constant}$$

5.4 The barometric formula

For an ideal gas, we have that

$$E \sim N \left(\frac{N}{V} \right)^{\frac{2}{3}} e^{\frac{2S}{3N}}$$

in the absence of external fields, so

$$\mu = \left(\frac{\partial E}{\partial N} \right)_{S,V} = \frac{E}{N} \left(\frac{5}{3} - \frac{2}{3} \frac{S}{N} \right) = \frac{5}{2}T - \frac{S}{N}T$$

In order for $\mu + U$ to be constant, as the gravitational potential energy increases, μ decreases. Plugging back in the Sackur-Tetrode equation,⁵ we see that

$$\mu = -T \ln \left(\frac{1}{(2\pi\hbar)^3} \frac{V}{N} \left(\frac{4\pi m E}{3N} \right)^{3/2} \right)$$

so

$$U(x) - T \ln \left(\frac{1}{(2\pi\hbar)^3} \frac{V}{N} \left(\frac{4\pi m E}{3N} \right)^{3/2} \right) = \text{const.}$$

From this, we are able to recover the *barometric formula*

$$\rho(x) := \frac{N}{V} \sim e^{-U(x)/T}$$

6 The Canonical Ensemble

6.1 Derivation

We're going to take a different look at thermodynamic systems. Imagine we have a system i weakly coupled to a large heat bath R at temperature T_{bath} such that the total energy $E_{\text{total}} = E_{\text{bath}} + E_i$ is constant. The system can exchange energy with the bath without changing the temperature. We can describe the total system with a microcanonical ensemble by first counting the microstates

$$\Omega(E_{\text{total}}) = \sum_{\text{states of } i} \Omega_R(E_{\text{total}} - E_i) = \sum_i e^{S_R(E_{\text{total}} - E_i)}$$

⁵I feel like there's an easier way to do this

Taylor expanding the entropy,

$$\approx \sum_i e^{S_R(E_{\text{total}}) - \frac{\partial S}{\partial E_{\text{total}}} \cdot E_i} = e^{S_R(E_{\text{total}})} \sum_i e^{-\beta E_i}$$

where $\beta := \frac{1}{T_{\text{bath}}}$. The microcanonical ensemble says all of the microstates for the total system are equally likely. Hence, we get a probability distribution for the states of i

$$\rho(E_i) = \frac{1}{Z} e^{-\beta E_i}$$

where

$$Z(\beta) = \sum_i e^{-\beta E_i}$$

is the *partition function*.⁶

6.2 Distribution of E

It may look like the most likely energy state for our system is 0, which is suspicious. However, the number of states per energy level $e^{S(E_i)}$ increases with energy. We get that the distribution of energies looks like

$$\rho_E = e^{S(E_i) - \beta E_i}$$

The maximum occurs when $\frac{\partial S}{\partial E_i} - \beta = 0$, which is the energy at which the temperature of the system equals the temperature of the bath, exactly as expected.

We can Taylor expand at the peak E_0 to get that

$$\begin{aligned} -\beta E + S(E) &\approx -\beta E_0 + S(E_0) + \frac{1}{2} \frac{\partial^2 S}{\partial E^2} \Big|_{E=E_0} (E - E_0)^2 \\ &= -\beta E_0 + S(E_0) - \frac{1}{2T^2 C_v} (E - E_0)^2 \end{aligned}$$

Therefore, the peak is approximately Gaussian

$$\sim e^{-\frac{(E-E_0)^2}{2T^2 C_v}}$$

Assuming the area far from the peak contributes negligibly, we can compute

$$Z = \int_0^\infty e^{-\beta E + S(E)} dE \approx \int_0^\infty e^{-\beta E_0 + S(E_0) - \frac{1}{2T^2 C_v} (E - E_0)^2} = e^{-\beta E_0 + S(E_0)} \sqrt{2\pi T^2 C_v}$$

⁶The Z stands for “zis is ze partition function”

Hence

$$\ln(Z) \approx -\frac{E}{T} + S(E) = -\frac{F}{T}$$

$$F = -T \ln(Z)$$

Just trust me that this last line is a real equality. From the line before, we see that

$$E = -\frac{\partial}{\partial \beta} \ln(Z) = -\frac{1}{Z} \frac{\partial}{\partial \beta} \sum_i e^{-\beta E_i} = \sum_i \rho(E_i) E_i$$

which is exactly as expected.

6.3 Entropy, revisited

Recall that

$$\begin{aligned} S &= -\frac{\partial F}{\partial T} = \ln(Z) + T \frac{\partial \ln(Z)}{\partial \beta} \frac{\partial \beta}{\partial T} \\ &= \ln(Z) - \beta \frac{\partial \ln(Z)}{\partial \beta} = \ln(Z) + \sum_i \rho_i \beta E_i \\ &= \ln(Z) + \sum_i \rho_i (-\ln(\rho_i) - \ln(Z)) \\ &= -\sum_i \rho_i \ln(\rho_i) \end{aligned}$$

This is exactly the *Shannon entropy* in information theory. Here we see the connection between the two disparate subjects. In fact, we've seen this before: in the microcanonical ensemble, $\rho_i = \frac{1}{\Omega}$ for every state, so $S = \ln(\Omega)$.

6.4 Examples

In the two-level system ensemble (2.4.1), we can first compute

$$Z = \sum_{\text{all states}} e^{-\beta \epsilon N_\epsilon} = (1 + e^{-\beta \epsilon})^N$$

From this, we can compute

$$\begin{aligned} E &= -\frac{\partial}{\partial \beta} \ln(Z) = -N \frac{\partial}{\partial \beta} \ln(1 + e^{-\beta \epsilon}) \\ &= \frac{N \epsilon}{1 + e^{\beta \epsilon}} \end{aligned}$$

Look how easy it was to compute the relationship between energy and temperature! We can also take a look at the harmonic oscillator ensemble (2.4.2) and see (adding back in zero-point energy and assuming $T \gg \hbar\omega$) that

$$Z = \sum_{\text{all states}} e^{-\beta E_i} = \left(\frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} \right)^N = \left(\frac{1}{e^{\beta\hbar\omega/2} - e^{-\beta\hbar\omega/2}} \right)^N = \left(\frac{1}{2 \sinh\left(\frac{\hbar\omega}{2T}\right)} \right)^N$$

$$F = TN \ln \left(2 \sinh \left(\frac{\hbar\omega}{2T} \right) \right)$$

$$E = \frac{N\hbar\omega}{2} \left(\frac{\cosh\left(\frac{\hbar\omega}{2T}\right)}{\sinh\left(\frac{\hbar\omega}{2T}\right)} \right) = N\hbar\omega \left(\frac{1}{2} + \frac{1}{e^{\beta\hbar\omega} - 1} \right)$$

As you can see, Z is much more than a normalization constant!

6.5 The Equipartition Theorem

Assume we have a *classical* Hamiltonian H . Then

Theorem 6.1. *The Equipartition Theorem:*

$$\left\langle q_i \frac{\partial H}{\partial q_j} \right\rangle = \left\langle p_i \frac{\partial H}{\partial p_j} \right\rangle = T \delta_{ij}$$

Proof. The corresponding “classical” partition function is

$$Z = \int e^{-\beta H(q_i, p_i)} \prod_{i=1}^N \frac{dq_i dp_i}{2\pi\hbar}$$

We can then compute

$$\begin{aligned} \left\langle p_i \frac{\partial H}{\partial p_j} \right\rangle &= \frac{\int p_i \frac{\partial H}{\partial p_j} e^{-\beta H} dq dp}{\int e^{-\beta H} dq dp} = \frac{\int p_i \left(-\frac{1}{\beta} \frac{\partial}{\partial p_j} e^{-\beta H} \right) dq dp}{\int e^{-\beta H} dq dp} \\ &= \frac{\int \frac{\partial}{\partial p_j} \left(-\frac{1}{\beta} p_i e^{-\beta H} \right) dq dp + \frac{1}{\beta} \frac{\partial p_i}{\partial p_j} \int e^{-\beta H} dq dp}{\int e^{-\beta H} dq dp} \end{aligned}$$

Assuming sufficiently nice boundary conditions, the left term in the numerator is 0. Hence, we simply end up with

$$\left\langle p_i \frac{\partial H}{\partial p_j} \right\rangle = \frac{1}{\beta} \frac{\partial p_i}{\partial p_j} = T \delta_{ij}$$

and likewise for q . □

Note that an immediate consequence of this is that if our Hamiltonian looks like

$$H = \frac{1}{2m} \sum (p_{xi}^2 + p_{yi}^2 + p_{zi}^2) + V(x_1, \dots, x_N)$$

(which is a lot of them) then

$$\begin{aligned} 3T &= \left\langle p_{xi} \frac{\partial H}{\partial p_{xi}} \right\rangle + \left\langle p_{yi} \frac{\partial H}{\partial p_{yi}} \right\rangle + \left\langle p_{zi} \frac{\partial H}{\partial p_{zi}} \right\rangle \\ &= \frac{1}{m} \langle p_x^2 + p_y^2 + p_z^2 \rangle = 2 \langle KE \rangle \end{aligned}$$

so there is a real relation between temperature and the average kinetic energy of each particle as eluded to in (2.3).

7 The Grand Canonical Ensemble

7.1 The grand canonical partition function

In the canonical ensemble, we imagined attaching the system up to a bath that was able to exchange energy with the system at constant T . The *grand canonical ensemble* lets us imagine our system now able to exchange *particles* with the bath at a constant μ .

We know from the microcanonical ensemble that the probability of finding the system with a given E_i and N_i is proportional to

$$\Omega(E_{bath}, N_{bath}) = e^{S_{bath}(E_{bath}, N_{bath})} = e^{S_{bath}(E_{total}, N_{total}) - E_i \frac{\partial S}{\partial E} - N_i \frac{\partial S}{\partial N}} \propto e^{-\beta(E_i - \mu N_i)}$$

In this way, we obtain the *grand canonical partition function*

$$\mathcal{Z} = \sum_{\text{microstates } i} e^{-\beta(E_i - \mu N_i)}$$

where the sum runs over all possible particle numbers *and* energies.

7.2 Fluctuations of E and N

By staring hard at 6.2 (or by computing $\langle \delta E^2 \rangle = \frac{\partial^2}{\partial \beta^2} \ln(Z)$) you can deduce that

$$\langle \delta E^2 \rangle = T^2 C_v$$

Since $C_v \sim N$, the standard deviation $\sqrt{\langle \delta E^2 \rangle} \sim \sqrt{N}$. Hence, it becomes negligible compared to N in the thermodynamic limit. Likewise,

$$\frac{\partial \ln(\mathcal{Z})}{\partial \mu} = \beta \langle N \rangle \quad \frac{\partial^2 \ln(\mathcal{Z})}{\partial^2 \mu} = \beta^2 (\langle N^2 \rangle - \langle N \rangle^2)$$

so

$$\langle (\delta N)^2 \rangle = T \left(\frac{\partial \langle N \rangle}{\partial \mu} \right)$$

The factor on the right is related to the compressibility of the system. I was told to read Landau and Lifshitz for this, but to save you from having to touch that godforsaken book,⁷ the relevant computation starts with

$$\left(\frac{\partial N}{\partial \mu} \right)_{T,V} = \left(\frac{\partial N}{\partial P} \right)_{T,V} \left(\frac{\partial P}{\partial \mu} \right)_{T,V}$$

Using a Maxwell relation and noting that $N = V f(P, T)$ is extensive,

$$\left(\frac{\partial P}{\partial \mu} \right)_{T,V} = \left(\frac{\partial N}{\partial V} \right)_{T,\mu} = \frac{N}{V}$$

and

$$\frac{N}{V} \left(\frac{\partial N}{\partial P} \right)_{T,V} = N \left(\frac{\partial}{\partial P} \frac{N}{V} \right)_{T,V} = N \left(\frac{\partial}{\partial P} \frac{N}{V} \right)_{T,N} = -\frac{N^2}{V^2} \left(\frac{\partial V}{\partial P} \right)_{T,N}$$

The second-to-last equality is subtle but convince yourself that it follows from $N = V f(P, T)$. We can now finally note that $\kappa = -\frac{1}{V} \left(\frac{\partial V}{\partial P} \right)_{T,N}$ is the isothermal compressibility of the system, so

$$\langle (\delta N)^2 \rangle = \frac{TN^2}{V} \kappa$$

8 More on the Canonical Ensemble

I'm looking back at my handwritten notes from a month ago and I have no idea why they're in the order that they are.

⁷I've heard it's actually a very good book to learn from. It's just that humans are naturally afraid of things they don't understand.

8.1 Ideal gas in the canonical ensemble

We can write down the partition function

$$\begin{aligned} Z &= \sum_i e^{-\frac{E_i}{T}} = \frac{1}{N!} \int e^{-\frac{1}{T} \sum_{n=1}^{3N} p_n^2 / 2m} \left(\prod_{n=1}^N \frac{d\vec{x}_n d\vec{p}_n}{(2\pi\hbar)^3} \right) \\ &= \frac{V^N}{N!} \frac{1}{(2\pi\hbar)^{3N}} \left(\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mT}} dp \right)^{3N} = \frac{V^N}{N!} \frac{(2\pi mT)^{3N/2}}{(2\pi\hbar)^{3N}} \end{aligned}$$

From this, we can compute the Helmholtz free energy

$$F = -T \ln(Z) \approx -TN \ln \left(\frac{V}{N} \frac{(2\pi mT)^{3/2}}{(2\pi\hbar)^3} \right)$$

From this we can compute $S = -\frac{\partial F}{\partial T}$ and rederive the Sackur-Tetrode formula. We can also compute $E = F + TS$, but there's an easier way.

8.2 Dulong-Petit and the Virial Theorem

It's an immediate consequence of equipartition that for an ideal gas

$$\langle E \rangle = \frac{1}{2} \left\langle \sum_{n=1}^{3N} p_n \frac{\partial H}{\partial p_n} \right\rangle = \frac{1}{2} \sum_{n=1}^{3N} \langle p_n \frac{\partial H}{\partial p_n} \rangle = \frac{3}{2} NT$$

For a free gas following a generic dispersion relation

$$\epsilon = c|p|^\delta$$

we have per particle that

$$3T = \left\langle p_x \frac{\partial H}{\partial x} + p_y \frac{\partial H}{\partial y} + p_z \frac{\partial H}{\partial z} \right\rangle = \langle \delta \epsilon \rangle$$

so the total energy is

$$\langle E \rangle = \frac{3}{\delta} NT$$

Now, consider a (non-relativistic) system with a potential

$$H = \sum \frac{p_a^2}{2m} + c \sum_{\langle ij \rangle} |\vec{x}_i - \vec{x}_j|^\delta$$

Then

$$\langle K \rangle = \frac{3N}{2}T \quad \langle V \rangle = \frac{3}{\delta} \sum_N \langle q \frac{\partial H}{\partial q} \rangle = \frac{3N}{\delta}T$$

For a generic solid, we imagine our atoms bound in harmonic oscillator potentials with $V \sim r^2$. In this case the total energy

$$\langle E \rangle = \langle K \rangle + \langle V \rangle = 3NT$$

This means that

$$C_v = 3N$$

and we recover the *Dulong-Petit law*. Similarly, for a gravitationally bound system with $\delta = -1$, we get

$$\langle K \rangle = \frac{3N}{2}T \quad \langle V \rangle = -3NT$$

In this way we recover a form of the *virial theorem*

$$\langle K \rangle = -\frac{1}{2} \langle V \rangle$$

Observe that energy *decreases* with temperature, so the system has a negative specific heat. Freaky!

8.3 Internal degrees of freedom

We've been considering our particles as, well, *particles*, in the sense that they have no internal structure. However, we can also consider an ideal gas of particles whose individual Hamiltonians look like

$$H = \frac{p^2}{2m} + \epsilon_i$$

where ϵ_i is the energy due to *internal* motion (rotations, vibrations, etc.) Our partition function then neatly splits into

$$Z = \sum_i e^{-\beta \Sigma \left(\frac{p^2}{2m} + \epsilon \right)} = Z_{\text{pointlike}} \cdot (Z_{\text{1 particle internal}})^N$$

Our specific heat then also splits

$$C_v = \beta^2 \frac{\partial^2}{\partial \beta^2} \ln(Z) = \frac{3}{2}N + Nc_v^{\text{int.}}$$

where $\frac{3}{2}N$ is the pointlike heat capacity. Let's explore some of the internal degrees of freedom of a typical molecule.

One source of energy comes from the *electronic* degrees of freedom. In principle, the atoms in the molecule could acquire enough energy to excite the electrons to a higher orbital. The energy associated with this transition is on the order of

$$E \sim \frac{e^2}{a_0} \sim \text{a few eV} \sim 10^4 K$$

which is very hot. Therefore, at everyday conditions, this degree of freedom is not activated.

Another source of energy is the *vibrational* degrees of freedom. The atoms in a molecule sit in a potential well that can be approximated by a QHO $V \approx \frac{k}{2}(R - R_0)^2$. Using esoteric quantum chemistry bullshit (but really just the Born-Oppenheimer approximation) we get

$$k \sim \frac{\hbar^2}{m_e a_B^4} \quad E_{\text{vib}} = \hbar\omega \sim \sqrt{\frac{\hbar}{M}} \frac{\hbar^2}{m_e a_B^2} \sim 1000 K$$

The last source is *rotational* degrees of freedom. We have that

$$E_{\text{rot}} = \frac{L^2}{2I} \sim \frac{\hbar^2}{Ma_B^2} \sim 1 K$$

This is therefore the only degree of freedom that is activated at everyday conditions. We see a rough hierarchy emerge

$$E_{\text{rot}} \ll_{\times \sqrt{\frac{m_e}{M}}} E_{\text{vib}} \ll_{\times \sqrt{\frac{m_e}{M}}} E_e$$

8.4 Heat capacities of rotating molecules

What do I mean by a degree of freedom being “activated”? Observe that the “internal partition function” for 1 molecule is given by

$$Z = \sum_L^{\infty} (2L+1) e^{-\beta T_{\text{rot}} L(L+1)}$$

In the low-temperature limit this becomes

$$Z \approx 1 + 3e^{-\beta T_{\text{rot}}}$$